

and control groups substantially exacerbates the selection problem in conducting evaluations.

14.2. *Matching results for the SIPP samples*

The column labelled “SIPP” in Table 6(a) presents the estimated bias from the local linear regression matching estimator based on the best-predicting model for $P(X)$.³⁰ A new model for $P(X)$ is fit using the SIPP data pooled with the data from the experimental control group. The data are coarser since we lack the detailed geographical location available in the ENP data. Otherwise, the model for $P(X)$ is very similar in specification to that used in the ENP-control analysis. The contribution of questionnaire and labour market mismatch to evaluation bias B is of the same order of magnitude as the contribution of poor predictors. LaLonde’s influential study used comparison groups that are mismatched both geographically and in terms of the questionnaires. Even closely regionally-aligned SIPP samples produce estimates that are substantially biased. Applying the difference-in-differences version of the estimator often attenuates the bias (see Table 6(b)) as would be expected if discrepancies in questionnaires and local labour markets are temporally-stable sources of the participant group—comparison group bias.

14.3. *An internal check of the importance of geographical mismatch*

In an effort to disentangle survey effects from local labour market effects, we split the JTPA ENP-experimental control data into two geographically-mismatched samples. Since the questionnaires are the same for both groups, the only new source of bias is local labour market or site effects. The sample splitting reduces the sample sizes, and the youth estimates can no longer be considered very reliable because the samples are so small. The induced bias is substantial (see Table 6(a) column labelled “Site Mismatch”), and comparable in absolute value to the bias that results from using SIPP. Note that the differences in estimated geographical effects across demographic groups rules out cost of living differences as the major reason for local labour markets effects. Presumably those effects would be the same across demographic groups. Taking conditional difference-in-differences greatly attenuates the bias, as would be expected if local labour market characteristics are persistent over time. (See Table 6(b).)

Smith (1995) investigates the separate roles of survey instruments and geographical mismatch in accounting for the discrepancy between ENP and SIPP earnings. He concludes that about two-thirds of the discrepancy between the two data sources is due to differences in the questionnaires and the rest is attributable to geographical mismatch of participants and control group members.³¹

15. USING NO-SHOWS AS A COMPARISON GROUP

For most social programmes, there is a group of persons who apply to the programme and are accepted into it but for some reason do not enroll in it. These people are distinct

30. A complete set of SIPP results in the format of Table 5 is available on request from the authors. The numbers in Table 6(a) and the numbers in Table 6(b) are based on the local linear regression estimator used in the third to last column and the last column of Tables 5, respectively. The bias estimates for the other estimators is of the same order of magnitude as the SIPP estimators recorded in Table 6(a) and 6(b).

31. His findings are of general interest because the JTPA data were collected in the format of the widely-used NLSY (National Longitudinal Survey of Youth) data.

from dropouts among enrollees. It may be that in learning about the programme through applying for it, they change their mind or that better opportunities arise.³² No-shows are in many ways similar to treated persons, and are good candidates for a comparison group. They are located in the same labour market as participants and are administered the same questionnaire as participants. If not enrolling in a programme were random with respect to outcomes, then non-enrollees would be like an experimental control group. In reality, enrollment is probably not random, but we can attempt to control for the differences between non-enrollees and treated persons using the same methods as used to compare programme participants with ENP comparison group members.

We assess the performance of matching methods for eliminating differences between no-shows and a randomized-out control group. As before, $D=1$ refers to controls but now $D=0$ refers to no-shows. In this context, $P(X)$ represents the conditional probability of being an enrollee (i.e. of not being a no-show) among persons who at one stage expressed their intention to enroll in the programme.³³ Figure 2 plots the empirical densities of estimated $P(X)$ values which look very similar across the control and non-enrollee groups. The bottom section of Table 2 reports the mean difference in outcomes between controls and no-shows, B , and its components for all four demographic groups. For three of four groups, the bias B is smaller than it is for the ENP—control comparison. However, the composition of B is weighted more heavily toward selection bias component B_3 . B is lower for adults than it is in the ENP-control comparison but the proportion of B due to selection bias is larger. For male youths, B is substantial and the contribution of selection bias B_3 is large both absolutely and proportionately.

Table 7 presents the estimated quarterly bias estimates associated with the local linear matching and regression-adjusted local linear matching estimators. The results from the other matching methods are comparable and are available on request from the authors. The first column of each table indicates that for the adult demographic groups, there is very little bias, as conventionally measured, (B), that arises from using a simple mean difference estimator. For the adult groups, the raw means for no-shows are very similar to those for the randomized-out control group without any adjustment for group characteristics. When matching methods are used, and a more theoretically appropriate measure of bias is used, the local linear regression-adjusted matching estimator emerges as an effective method but in general the matching estimators yield biases that are slightly higher than the difference in raw means. These results are not surprising. The components of bias B discussed in Section 9, may coincidentally offset each other. Small values of B do not imply a small bias for $M(S)$. Data limitations prevent us from applying the conditional difference-in-differences estimator to the no-show data.³⁴

In the adult samples, it appears that the no-shows are already well-matched to the control group as indicated by the densities of $P(X)$, so matching would not be expected to lead to a significant improvement over the raw means. However, the discrepancy between $P(X)$ for $D=1$ and $D=0$ is greater for youth and the performance of matching is worse for the youth samples.³⁵

32. Still another possibility considered in the literature on performance standards is that programme administrators have an incentive to give persons trial treatments before enrolling them in the programme. Through such interventions, they can learn about the probable success of the person in the programme and then encourage or discourage that individual from enrolling. Individuals who fail to enroll are usually not counted against the JTPA site. There is some evidence that this type of behaviour, called cream-skimming, took place at JTPA sites. (See Heckman, Smith and Taber (1996).)

33. Estimated coefficients are reported in Appendix E, which is available on request from the authors.

34. Pre-programme earnings data are not available for JTPA no-shows.

35. We observe lower prediction rates for the no-show model than for the participation model, which indicates that we are better able to predict the programme participation decision than the no-show decision. This is due in part to the way we collected the data on programme participation which focused on the application-acceptance decision (ENP-Control).

TABLE 7
*Estimated bias for alternative nonparametric matching methods**
Experimental controls and no-shows†

Quarter	Difference in means	Local linear P score matching	Regression-adjusted local linear matching	Difference in means	Regression-adjusted local linear matching	
					Local linear P score matching	Difference in means
Adult males						
$t = 1$	64 (35)	66 (39)	58 (38)	17 (24)	28 (30)	26 (28)
$t = 2$	32 (37)	45 (40)	37 (39)	7 (30)	11 (38)	9 (36)
$t = 3$	26 (41)	36 (42)	27 (42)	-5 (38)	-9 (38)	-13 (37)
$t = 4$	19 (46)	3 (49)	-6 (48)	12 (28)	5 (36)	2 (35)
$t = 5$	22 (49)	10 (51)	1 (48)	14 (24)	4 (32)	1 (31)
$t = 6$	7 (50)	-12 (52)	-21 (48)	11 (27)	1 (36)	-2 (34)
Ave. 1 to 6	29 (37)	25 (39)	16 (37)	9 (23)	7 (31)	4 (30)
As a % of impact**	66%	57%	37%	32%	23%	14%
As a % of adjusted impact	47%	41%	26%	27%	20%	11%
As a % of Control-ENP	8%	53%	42%	29%	18%	10%
Female youth						
Male youth						
$t = 1$	92 (37)	116 (41)	104 (44)	12 (30)	53 (33)	55 (32)
$t = 2$	33 (38)	49 (43)	36 (43)	17 (27)	52 (37)	52 (32)
$t = 3$	56 (42)	80 (47)	70 (48)	39 (31)	80 (35)	74 (34)
$t = 4$	111 (36)	133 (42)	116 (45)	-7 (32)	23 (39)	21 (36)
$t = 5$	100 (39)	108 (46)	95 (48)	17 (34)	39 (46)	37 (39)
$t = 6$	111 (43)	111 (47)	108 (53)	30 (31)	58 (39)	57 (36)
Ave. 1 to 6	84 (31)	99 (34)	88 (38)	18 (23)	51 (30)	49 (26)
As a % of impact**	144%	171%	152%	2639%	7474%	7273%
As a % of adjusted impact	233%	276%	245%	73%	207%	201%
As a % of Control-ENP	419%	255%	1258%	37%	83%	618%

* The table reports the bias for alternative matching methods. The bias in the first column is $\hat{\beta}$. The estimators in the second and third columns for each group restrict matches to the common support region and the bias estimates correspond to $\hat{B}_{SP}$.

† The predictor model given in the second footnote to Table 2 is used to estimate the probability of programme participation. The conditioning variables in the regression adjusted local linear models are site, age, education, race, the local unemployment rate, indicator variables for marital status and for the presence of children aged less than 6 in the household, and indicators for the quarter of the year and the year.

‡ A 2% trimming rule is used to determine the region of overlapping support for adult groups and a 5% trimming rule is used for youth groups (see Appendix C). A fixed bandwidth equal to 0.06 and a biweight kernel, defined in Appendix A, are used for nonparametric estimates.

** The impacts in the table are mean monthly impacts for the six post-programme quarters, estimated using the experimental treatment and control data for the four JTPA training sites in our study. See the experimental impacts and the adjusted impacts in Table 4.

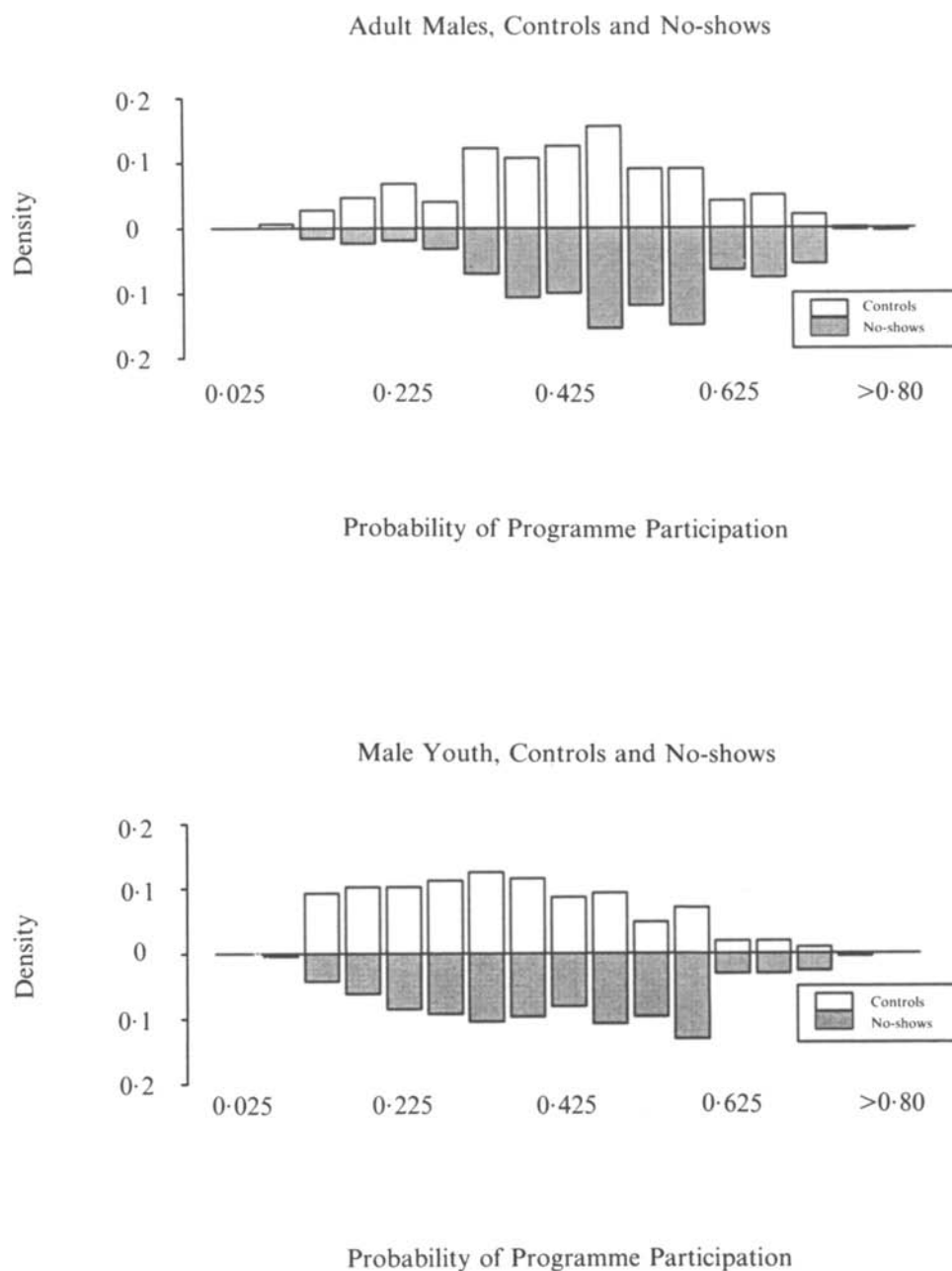


FIGURE 2. Density of Estimated Probability of Programme Participation for No-shows